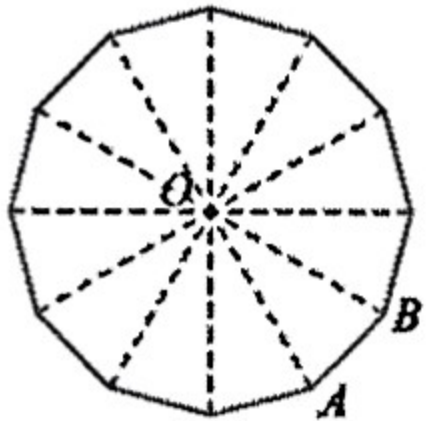
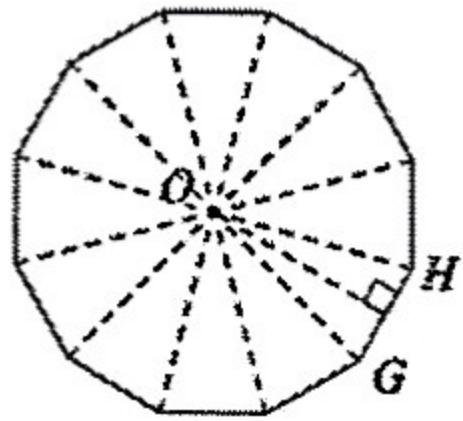


Inequalities WS 4

1. Suppose $a > 0, b > 0$ and $c > 0$
 - a. Prove $a^2 + b^2 \geq 2ab$
 - b. Hence prove $a^2 + b^2 + c^2 \geq ab + bc + ca$
 - c. Prove that $a^3 + b^3 + c^3 \geq 3abc$
 - d. If x, y, z are > 0 , Make a suitable substitution into (c) to show that $x + y + z \geq 3(xyz)^{\frac{1}{3}}$
 - e. Suppose $(1+x)(1+y)(1+z) = 8$, prove that $xyz \leq 1$
2. a. Given that x and y are non-negative, prove that $\frac{x+y}{2} \geq \sqrt{xy}$.
 - b. Hence prove that $(x+y)(x+z)(y+z) \geq 8xyz$, where z is also non-negative.
3. a. Show that $a^2 + b^2 > -2ab$ for all real numbers a and b .
 - b. Hence deduce that for all positive real numbers a, b and c :
 - i. $(a+b+c)^2 \geq 3(ab+bc+ca)$
 - ii. $ab(a+b) + bc(b+c) + ca(c+a) \geq 6abc$
 - c. Suppose that a, b and c are the side lengths of a triangle.
 - i. Explain why $(b-c)^2 \leq a^2$
 - ii. Deduce that $(a+b+c)^2 \leq 4(ab+bc+ca)$
4. a. A regular dodecagon is drawn inside a circle of radius 1 cm and centre O so that its vertices lie on the circumference. Determine the area of $\triangle OAB$, and hence find the exact area of the inscribed dodecagon.
 
 - b. i. Use the formula for $\tan 2\theta$ to show that $\tan 15^\circ = 2 - \sqrt{3}$.
 - ii. Another regular dodecagon is drawn with centre O , so that each side is tangent to the circle. Find the area of $\triangle OGH$ and hence find the exact area of the circumscribed dodecagon.
 
 - iii. By considering the results in parts (a) and (b), show that $3 < \pi < 12(2 - \sqrt{3}) \div 3.24$.
5. a. If $a > 0, b > 0$ show that $\frac{a+b}{2} \geq \sqrt{ab}$
 - b. If $a > 0, b > 0$ show that $\frac{a+b}{2} \leq \sqrt{\frac{a^2+b^2}{2}}$
6. A rectangular box of sides a, b, c has a constant surface area S .

Using the formulas $S = 2(ab + bc + ca)$ and $V = abc$ (volume), prove that:

 - a. $V^{\frac{2}{3}} \leq \frac{S}{6}$
 - b. V is a maximum when the box is a cube.

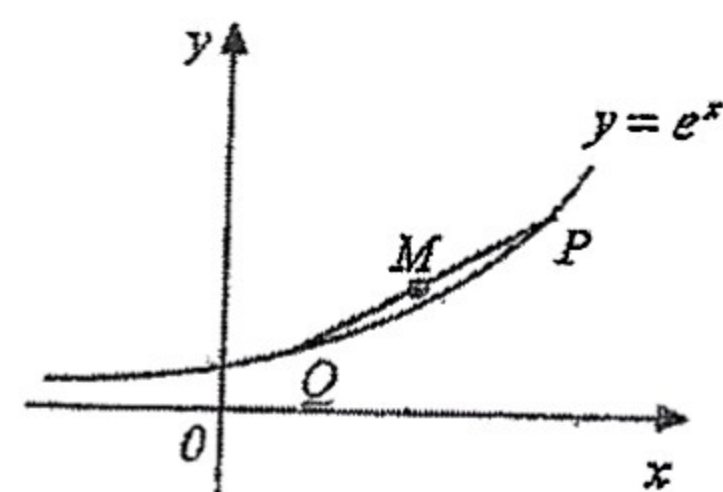
7. $P(a, e^a)$ and $Q(b, e^b)$, where $a > b$, are two points on the curve $y = e^x$.

M is the midpoint of PQ .

a. Use the diagram to show that $e^a + e^b > 2e^{\frac{1}{2}(a+b)}$

b. Hence show that if $a > b > c > d$

$$\text{then } e^a + e^b + e^c + e^d > 4e^{\frac{1}{4}(a+b+c+d)}$$

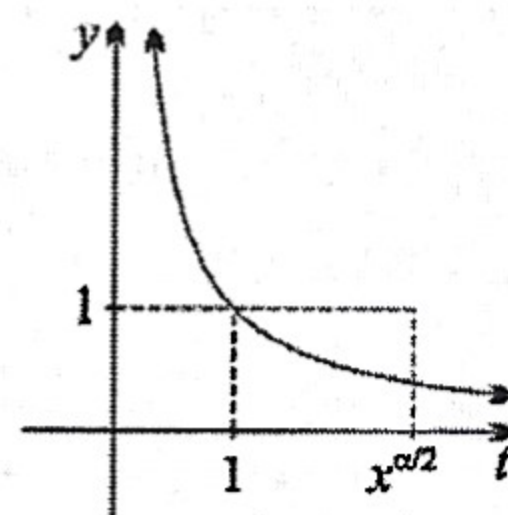


8. The diagram shows the graph of $y = \frac{1}{t}$, for $t > 0$. Let $x > 1$ and $\alpha < 0$.

a. By considering upper and lower rectangles, show that

$$0 < \frac{1}{2}\alpha \log x < x^{\frac{\alpha}{2}}.$$

b. Hence show that $\lim_{x \rightarrow \infty} \left(\frac{\log x}{x^\alpha} \right) = 0$, for all $\alpha > 0$.



9. Show that $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$

a. Hence show that

$$\tan^2 \left(\frac{\alpha + \beta}{2} \right) - \tan \alpha \tan \beta = \frac{\cos(\alpha + \beta)(1 - \cos(\alpha - \beta))}{\cos \alpha \cos \beta (1 + \cos(\alpha + \beta))}$$

b. Hence show that for $0 < \alpha < \frac{\pi}{4}$ and $0 < \beta < \frac{\pi}{4}$, $\sqrt{\tan \alpha \tan \beta} \leq \tan \left(\frac{\alpha + \beta}{2} \right)$.

10. Let $|t| < 1$ let N be a positive integer.

a. Show that $1 + t^2 + t^4 + \dots + t^{2N} < \frac{1}{1-t^2}$.

b. Show that the difference between the two is $\frac{t^{2N+2}}{1-t^2}$.

c. Integrate the results in part (a) between 0 and x , where $|x| < 1$. Hence show that:

$$x + \frac{x^3}{3} + \frac{x^5}{5} + \dots + \frac{x^{2N+1}}{2N+1} < \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right).$$

11. It is given that if a, b, c are any three positive real numbers, then $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$. If $a > 0, b > 0, c > 0$

are real numbers such that $a + b + c = 1$, use the given result to show that

a. $\frac{1}{abc} \geq 27$

b. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$

c. $(1-a)(1-b)(1-c) \geq 8abc$

12. Given $(x + y + z) \geq 3(xyz)^{\frac{1}{3}}$ for all x, y, z non-negative real numbers

a. Show that $xy + yz + zx \geq 3(xyz)^{\frac{2}{3}}$

b. Hence show that $xyz \leq (a-1)^3$ if $1 + (z + y + z) + (xy + xz + yz) + xyz = a^3$ where

$a \geq 2$ is a real number

Answer

3. a. See Question 1 (a); b. i. Expand the LHS and use part (a); c. i. The triangle inequality: the length of any side is between the sum of the other two and the difference of the other two; c.ii. Begin with LHS-RHS and use part (i).

4. a. $|\Delta OAB| = \frac{1}{4}$ sq. units, 3sq. units; b. (ii) $|\Delta OGH| = (2 - \sqrt{3})$ sq. units, $12(2 - \sqrt{3})$ sq. units